VASTHERA

Rewriting Redox, Restoring Life

VASTHERA discovers and develops first-in-class small molecule medicines that restore abnormal signaling pathways, transforming the lives of patients with rare and intractable diseases.

Precision Targeting

Pathway Restoration

Small Molecule Excellence

Impacting Lives with Hope

NORMALIZING SIGNALING PATHWAYS
Correcting the root causes of disease

RESTORING CELLULAR BALANCE
Re-establishing healthy cellular function

TRANSFORMING LIVES
Delivering new hope and better outcomes

SCIENCE. PRECISION. COMPASSION. | FOR PATIENTS. FOR FAMILIES. FOR THE FUTURE.

Disclaimer

This presentation is for informational purposes only and does not constitute an offer, commitment, or binding agreement. It contains forward-looking statements that involve risks and uncertainties, and actual results may differ materially. No representation or warranty, express or implied, is made regarding the accuracy, completeness, or reliability of the information provided. This document may include unpublished data and proprietary insights. It is intended solely for the recipient and should not be disclosed, distributed, or used for any purpose other than evaluating a potential partnership. VASTHERA assumes no liability for reliance on this information.

VASTHERA

Company Profile

About VASTHERA

VASTHERA Co., Ltd. is a pharmaceutical company developing first-in-class small molecule drugs through its proprietary drug discovery platform Redoxizyme™ to treat cardiovascular, neurodegenerative, and oncologic disorders.

80% R&D employees

40 ea Global IPs

3 Assets First-in-Class Therapeutics

Government & Public Grants

~12.6M USD Secured

Our Mission

VASTHERA aims to develop innovative drugs that 'cure' diseases rather than simply 'alleviate' them, ultimately benefiting patients.

Location: Seoul, South Korea

Members: 17 employees / 3 SAB

Technology Platform: Redoxizyme™

Company History

2018	Foundation
2020	Series A 3.7M USD
2022	Series B 12.2M USD
2022	Received Commendation Award from the Minister of SMEs and Startups
2024	ODD granted to VTB-10 by FDA
2025	IND clearance for Phase 1 trial of VTB-10 from FDA
2026	First human dosed in Phase 1 trials
2026	Selected for KDDF funding

VASTHERA

Our Pipeline

Pipeline	Indication	Discovery	Lead Optimization	Pre-Clinical	Phase 1	Note
VTB-10	PAH			KDDF	KDDF	▪ ODD granted by FDA in 2024 ▪ IND filing in June 2025 ▪ Phase 1 started in 1Q of 2026 ▪ other CVD indications
VTA-27	AD			KDRC		▪ GLP-Tox started in 4Q of 2025 ▪ Potential expansion to PD
VTA-63	TNBC		KDDF			▪ GLP-Tox started in 4Q of 2025 ▪ Potential expansion to other cancers

Executive Leadership

VASTHERA's leadership team unites expert scientists and industry veterans with extensive experience in redox biology, medicinal chemistry, medical science, and cardiovascular research from leading institutions in the Republic of Korea and the United States.

Sang Won Kang, PhD

CEO and Founder

Professor at Ewha Woman's University, and former CSO at LabFrontier, with careers in redox biology spanning academia, biotech, and NIH.

Kyung Joo Lee, PhD

CTO

20+ years in pharma and biotech, specializing in medicinal chemistry, with senior R&D roles in South Korea and the US, including Scripps Research Institute.

Hyun Gyu Lee, MD, PhD

CDO

10 years+ of experience as a venture capitalist specializing in Healthcare and Biotech, with senior leadership roles including VP and COO positions at multiple Biotech companies.

Dong Hoon Kang, PhD

CSO

Medical science expert, Research Professor at Ewha Woman's University and Asan Medical Center. Awarded the President's Postdoctoral Fellowship.

Choong Hee Lee, MSc

Director of Clinical Development

17+ years in global and local clinical trials across pharma, CROs, and biotech, with experience in Clinical Development and Operation for Phase 1-4 study, and for Medicines and Healthcare products.

VASTHERA

VTA-27

For Every Memory That Matters

A First-in-Class Oral Therapeutic for
Alzheimer's Disease (AD)

2026

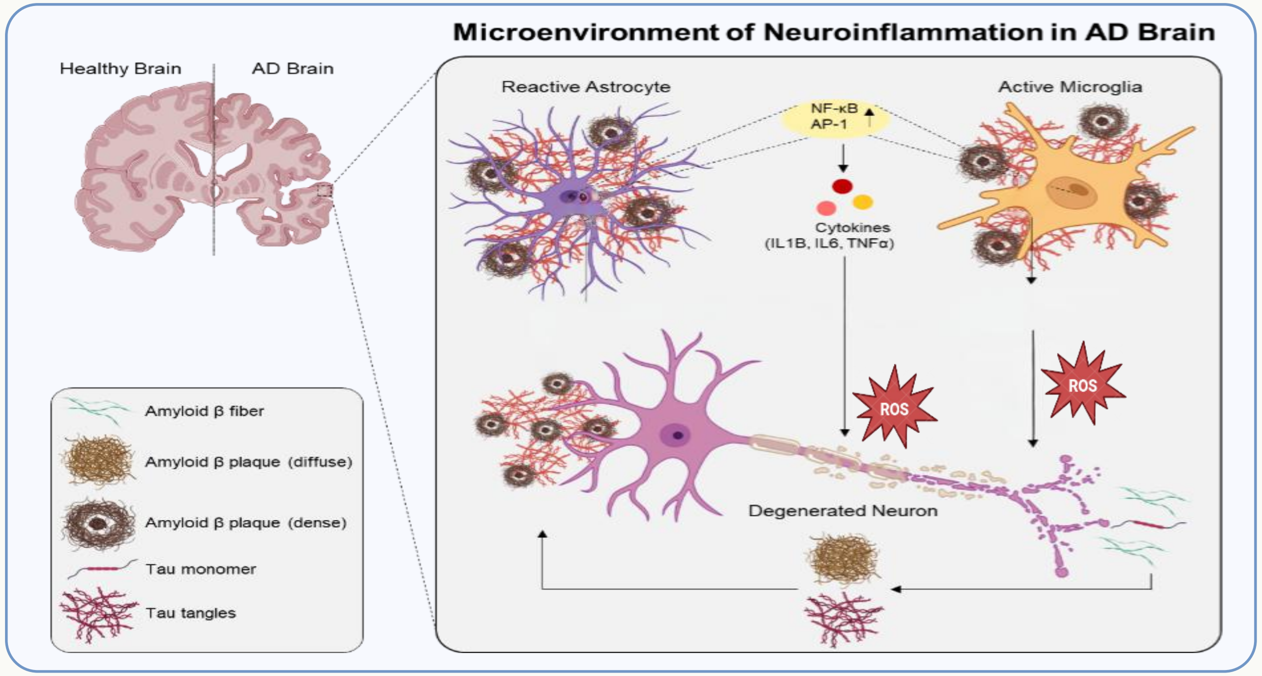
The Unmet Need

AD remains a progressive, life-limiting disease with high unmet medical need

VASTHERA

Neuroinflammation: A Critical Link Between Disease Severity and Cognitive Loss

1 GLIAL CELL BIOLOGY



2 FROM PATHOLOGY

Amyloid Plaques
Drive neurotoxicity and inflammation

Tau pathology
Disrupts microtubules and transport

Neuroinflammation
Sustained activation of innate immune response

Neuronal Stress
Energetic and proteostatic stress in neurons

BBB Dysfunction
Leads to leakage and immune cell infiltration

3 LEADING TO COGNITIVE DECLINE

Synaptic loss & network failure
Disconnectivity and neuronal dysfunction

Cognitive Decline
Impaired memory, executive function, & daily function

4 EMERGING EVIDENCE IMPLICATES GLIAL PATHWAYS

GLIAL ASSOCIATED TARGETS

- **APOE:** Lipid transport and astrocyte function
- **TREM2:** Microglial activation and survival
- **TAM (Tyro3/AXL/MER TK):** Phagocytosis and resolution of inflammation
- **Astrocytic STAT3:** Astrocyte reactivity and neuroprotection

TRANSLATIONAL SUPPORT

- Glial activation correlates with disease progression and cognitive decline
- Glial resilience associates with slower decline and resilience
- Preclinical models show that restoring glial function improves synaptic health and cognition

KEY MESSAGE

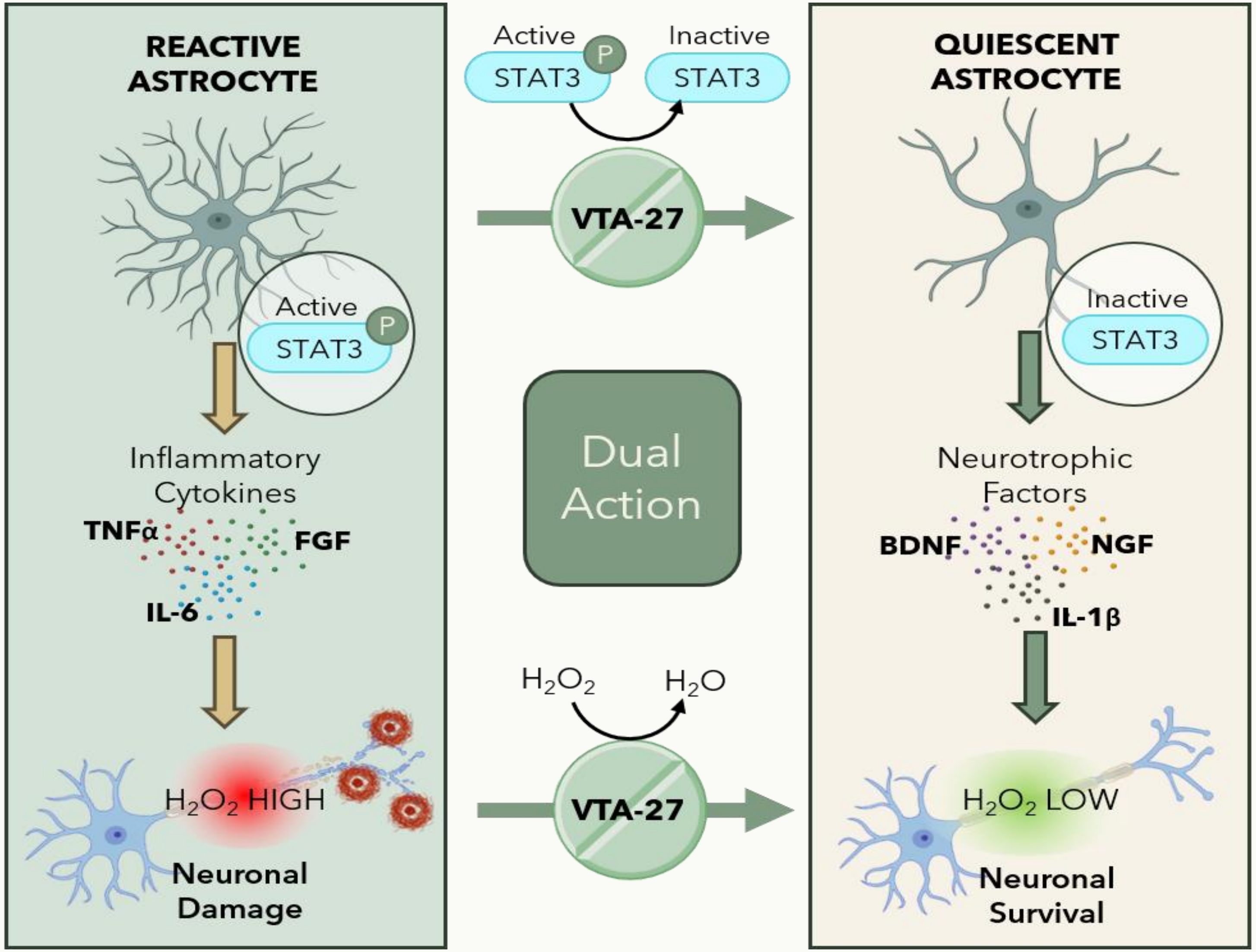
Astrocyte-targeted therapies go beyond amyloid clearance by restoring the neuro-immune axis essential for neuronal network and cognitive function.

VASTHERA

VTA-27: A First-in-Class Neuroinflammation Targeting Therapy

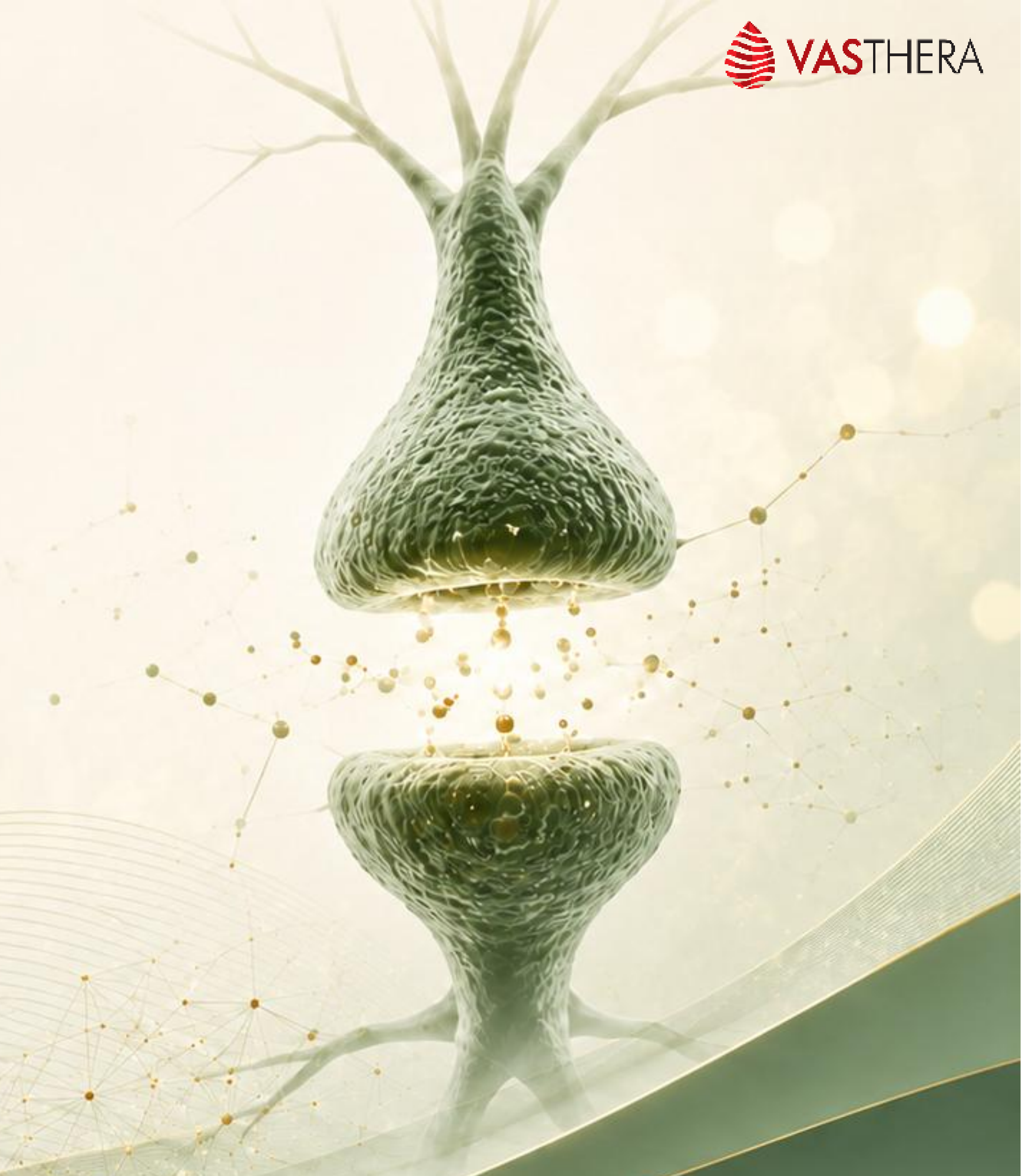
KEY MESSAGE

VTA-27 is designed to restore cognitive function by simultaneously shutting down reactive astrocytes and protecting neurons, offering a differentiated neuroinflammation targeting approach for Alzheimer's disease.



A Novel Disease-Modifying Mechanism

VTA-27 targets reactive astrocytes for the treatment of Alzheimer's Disease.



Peroxiredoxin: A Novel Therapeutic Target

1 BIOLOGICAL RATIONALE

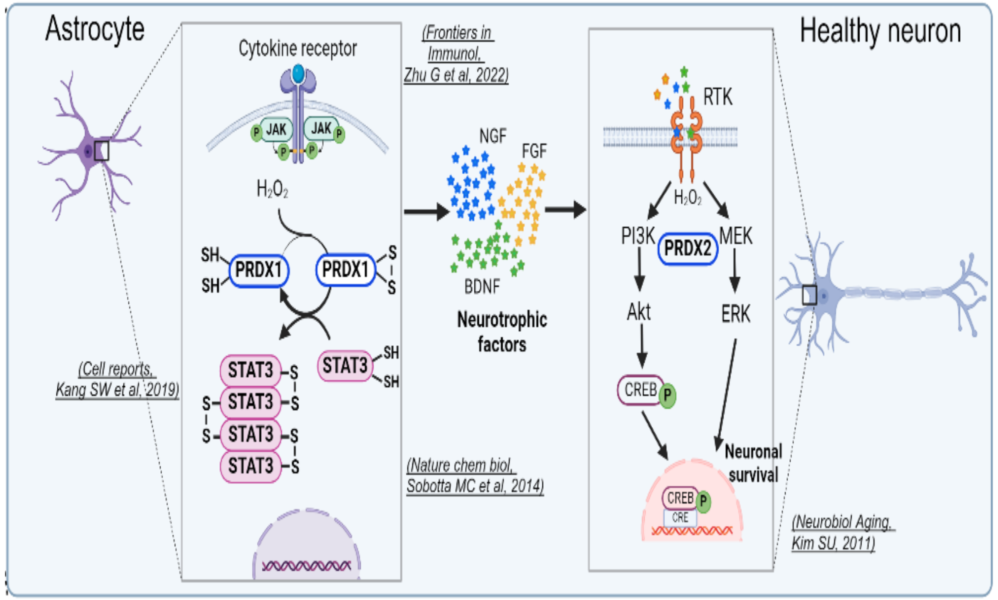
Peroxiredoxin: The Guardian of Redox Homeostasis

(PRX, gene name: PRDX)

Peroxiredoxins are thiol peroxidase enzymes that regulate cellular H_2O_2 as a catalytic substrate. They are essential for all aerobic organisms by tuning cellular redox signaling as well as further protecting cells from oxidative stress. They are the most sensitive enzyme against cellular H_2O_2 , thus described as the "Redox guardian".

CELL-TYPE SPECIFIC FUNCTION OF PRX1 AND PRX2 IN THE BRAIN

In the brain, PRX1 isoform is abundantly expressed in astrocytes, while PRX2 isoform is expressed selectively in neurons. Thus, both isoforms are believed to be involved in the astrocyte-neuron communication network essential for memory and cognitive functions.



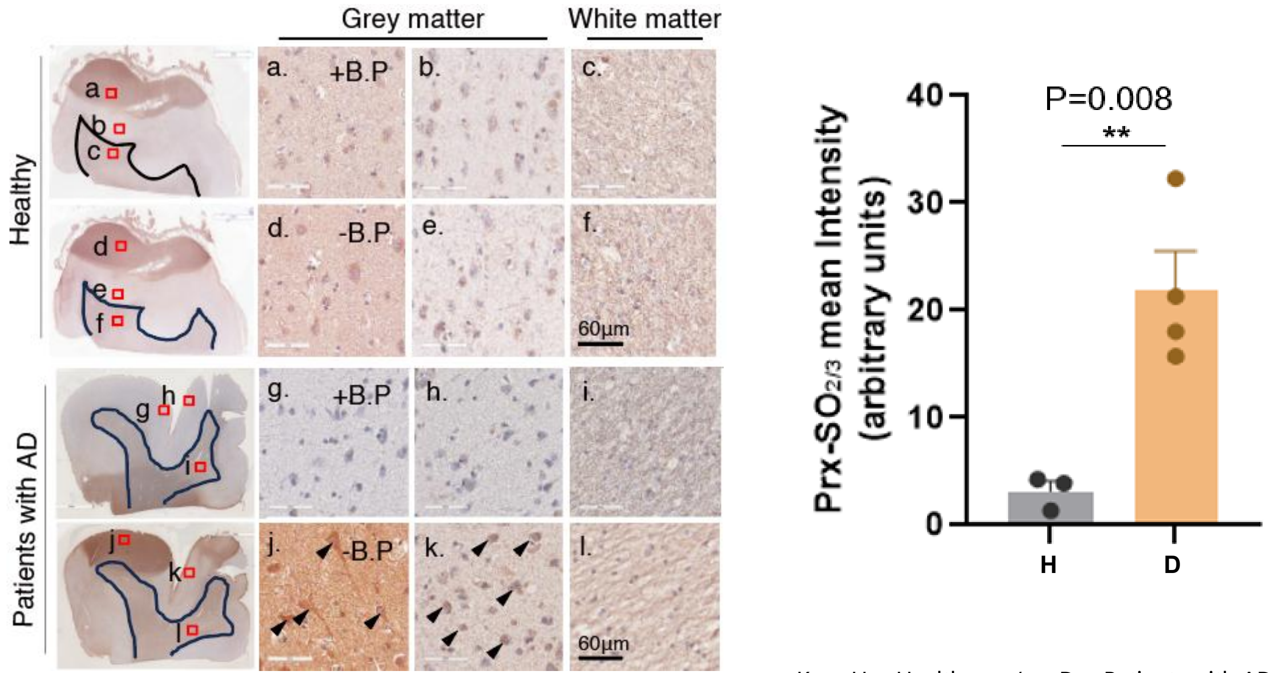
2 IMMUNOSTAINING FOR HYPEROXIDIZED PRX (DEAD ENZYME FORM)

STUDY DESIGN

Post-mortem brain tissues from AD patients (n = 4) and age-matched healthy controls (n = 3) were stained for hyperoxidized PRX (PRX-$SO_2/_3$) – the catalytically inactive form of the enzyme – by immunohistochemistry. Grey and white matter were assessed separately, antibody specificity was confirmed with a blocking-peptide control (+B.P / –B.P), and grey-matter signal intensity was quantified and compared between groups.

STUDY RESULTS

Compared to healthy brains, patients with AD showed increased level of PRX hyperoxidation. This suggests a possibility that PRX inactivation may exacerbate the pathogenesis of AD.



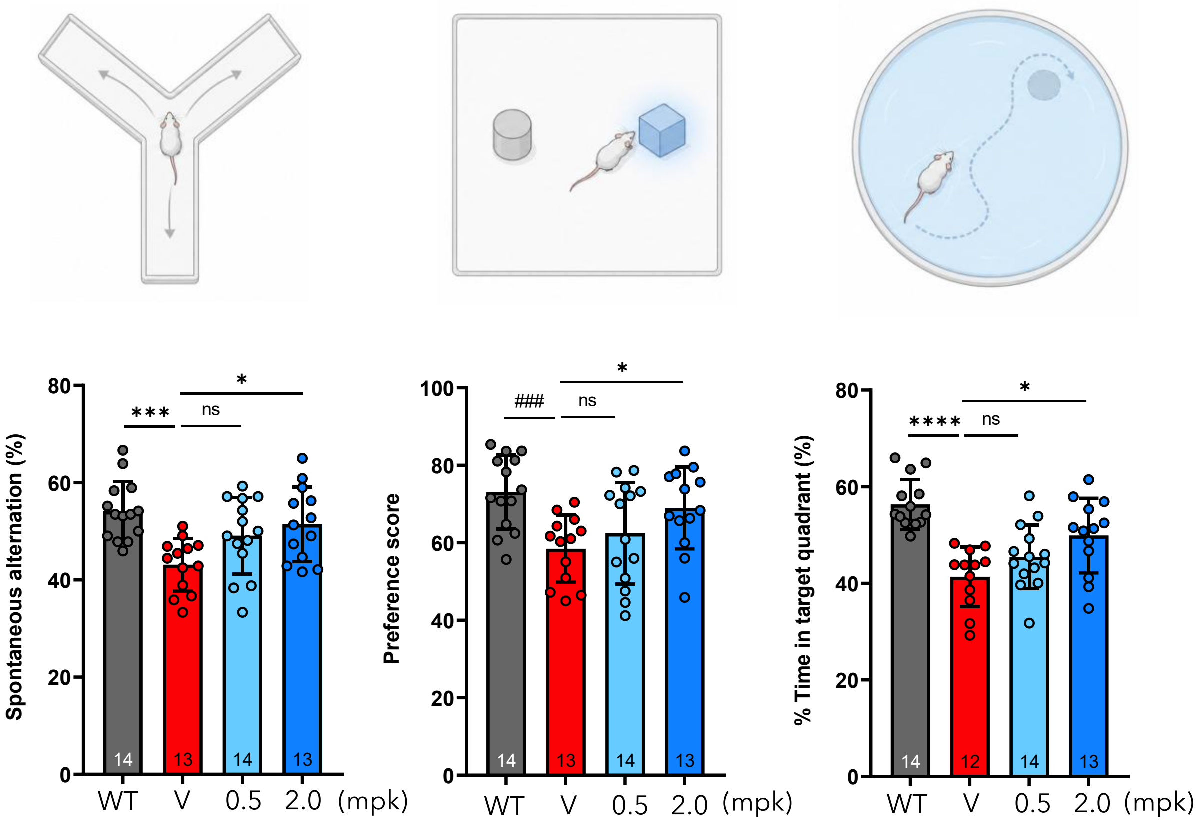
Key: H = Healthy / D = Patients with AD

VTA-27 Restores Cognitive Impairment

Y-MAZE

NOR
Novel Object Recognition

MWM
Morris Water Maze



STUDY DESIGN

Following 8 weeks of VTA-27 treatment, cognition-related behavioral performance was assessed using Y-maze, NOR, and MWM tests, covering working memory, recognition memory, and spatial learning/memory.

STUDY RESULTS

Vehicle-treated 5xFAD mice showed impaired performance across cognition-related behavioral assays. VTA-27 showed a recovery trend at 0.5 mg/kg and more consistent improvement at 2 mg/kg across Y-maze, NOR, and MWM, supporting functional benefits across multiple memory domains.

KEY MESSAGE

VTA-27 improves cognition-related behavioral performance in 5xFAD mice, supporting functional translation of its hippocampal neuroprotective effects.

Beyond Amyloid:
VTA-27 Addresses the Key Downstream Drivers of Alzheimer's Disease

	Neural Network Restoration Restores synapses And neural circuits	Cognitive Recovery Improves learning, memory and function	Anti-Neuroinflammation Restores chronic neuroinflammation	Vascular Restoration Restores cerebral vascular health	Amyloid Plaque Removal Reduces amyloid plaques
Leqembi / Kinsula (anti-A)					●
BIIB080 (anti-tau)	●	●			
TREM2 / NLRP3		●	●		
Neurotrophic (BDNF, NGF)	●	●			
VTA-27	●	●	●	●	● At high dose

VTA-27 can be a multi-target therapeutic by uniquely eliciting neuronal restoration, cognitive recovery, anti-inflammatory activity, and vascular regeneration. Addressing key downstream drivers beyond amyloid to support meaningful clinical benefit

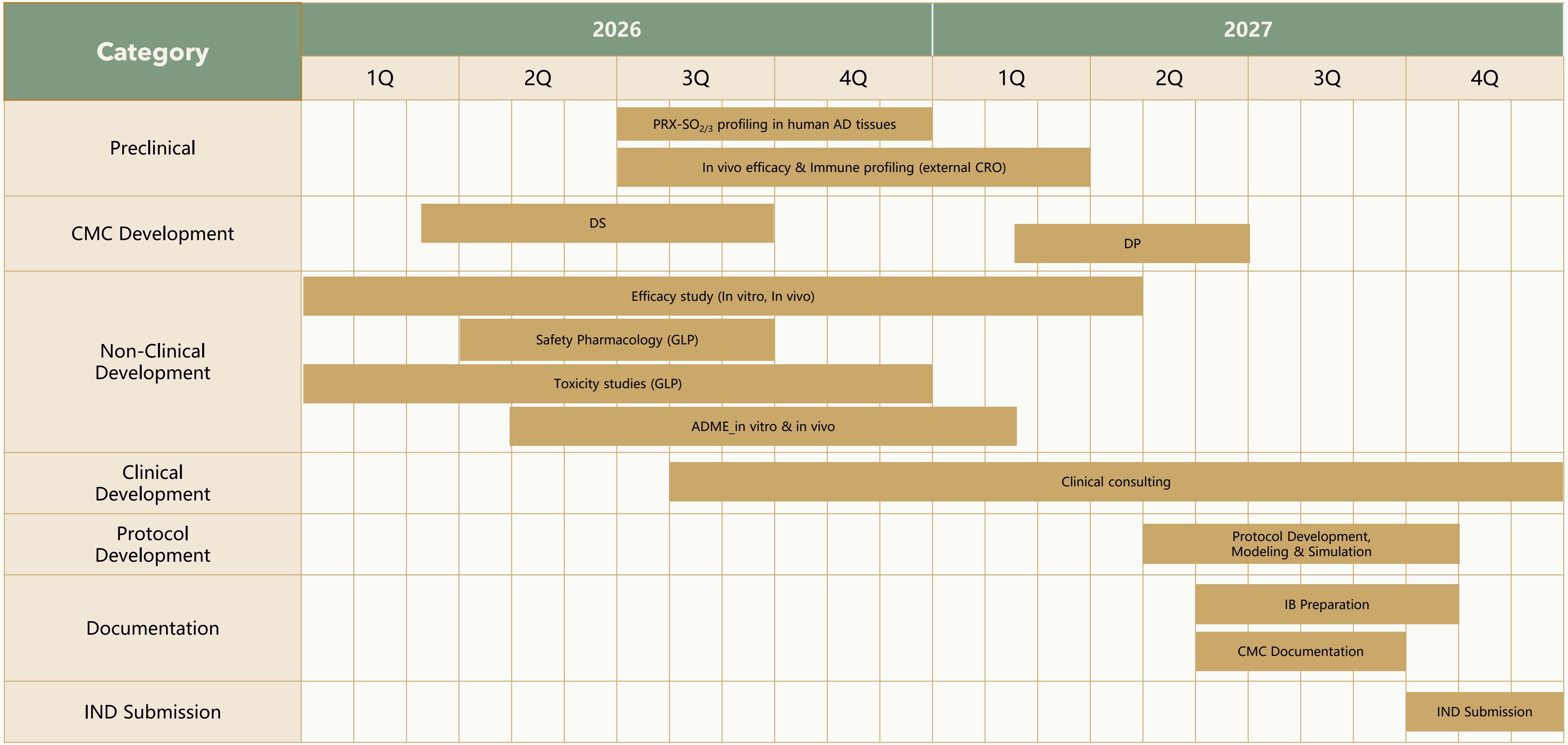
IND-Enabling Study

- Robust safety and toxicology package
- Pharmacology and PK profile support once-daily oral dosing
- Manufacturing process and CMC

Candidate Profile

Items	Contents
Indication	Alzheimer's Disease (AD)
Target Population	Mild to moderate AD patients
Substance	Small-molecule compound (VTA-27)
Administration route and frequency	**Oral** administration, **once daily**
MoA	VTA-27 inhibits astrocyte reactivation through redox-modulation of STAT3 activity
In vitro efficacy	Inhibition of STAT3 phosphorylation in primary astrocytes (IC_{50} = 162nM)
In vivo efficacy	Short working memory improvement at 2mg /kg dose (i.p. twice weekly for 8 weeks) in 5xFAD mouse Verified by external CRO: Y maze, Novel object recognition, Morris water maze
DMPK	**ADME:** in vivo PK (mouse, rat, dog & monkey), rat distribution and rat excretion as well as in vitro ADME are underway
Toxicity	**GLP Tox study:** 28-day repeat-dose toxicology studies completed in both rat and dog models. **Genotoxicity & Safety Pharmacology:** Ames test is negative. Safety pharmacology battery is clear, including hERG (IC_{50}>200 μM).
CMC	Synthetic route optimized with high purity (>99%) and low total impurities (<0.5%). - DS (GMP) production is underway - DP (GMP, active & placebo tablets) manufacturing is planned
Intellectual property	Material IP (PCT) ready to file

Development Timeline for VTA-27



Contact Us

Phone
+82-2-3277-4908

Mail
bd@vasthera.com

Website
www.vasthera.com

Headquarters
#504, Science Building C,
52, Ewhayeodae-gil, Seodaemun-gu, Seoul, Korea

Research institute
#251, University-Industry Cooperation Building,
150, Bugahyeon-ro, Seodaemun-gu, Seoul, Korea